Simulating Mechanical Energy Transfer in Rowing

A Simplified One-Dimensional Model

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Abstract

This paper uses a simple model to examine how periodic thrust and quadratic drag affect a rowing shell's speed. A prescribed square-wave force accelerates a single mass during the drive and vanishes during recovery. The results illustrate speed fluctuations and energy accounting during a 20-second startup. The calculation does not measure rower efficiency or validate performance against experimental data.

1 Model

The boat and crew are represented by a single mass m with speed v. The equation of motion is

$$m\frac{dv}{dt} = F(t) - kv^2. \tag{1}$$

For the baseline, $m = 766.5$ kg, $k = 40$ kg/m, and the prescribed thrust is 3000 N for 0.9 seconds in each 2-second cycle, followed by zero thrust. The boat starts from rest. These values are illustrative model inputs, not measurements from a crew.

The model calculates thrust work $W = \int Fv\,dt$, drag work $D = \int kv^3\,dt$ and the change in kinetic energy $\Delta K = m(v_f^2 - v_0^2)/2$. In the continuous model $W - D = \Delta K$. Thrust work on the modeled mass is not the rowers' total mechanical or metabolic work. The ratio $\Delta K/W$ describes how much input work remains as kinetic energy at the end of this startup interval; it is not rowing efficiency and depends on the interval chosen.

2 Method and results

The equation is integrated with explicit Euler steps of 0.01 seconds for 20 seconds. The baseline average speed over the latter half is 5.785 m/s and the maximum is 6.759 m/s. Thrust work is approximately 141.2 kJ, drag work 132.3 kJ and kinetic energy gain 9.17 kJ. The gain is 6.49% of thrust work. The difference of about -316 J in $W - D - \Delta K$ is a numerical residual from the discrete scheme.

At a 0.005-second step the residual is about -158 J; at 0.001 seconds it is about -31 J. The corresponding latter-half average speeds are 5.782 and 5.781 m/s. This convergence check addresses discretization, not the physical accuracy of the inputs.

3 Parameter comparisons and limitations

With the default step, halving drag raises the latter-half average speed to 8.162 m/s; halving mass gives 5.719 m/s over the same finite interval. Shortening the cycle from 2.0 to 1.5 seconds gives

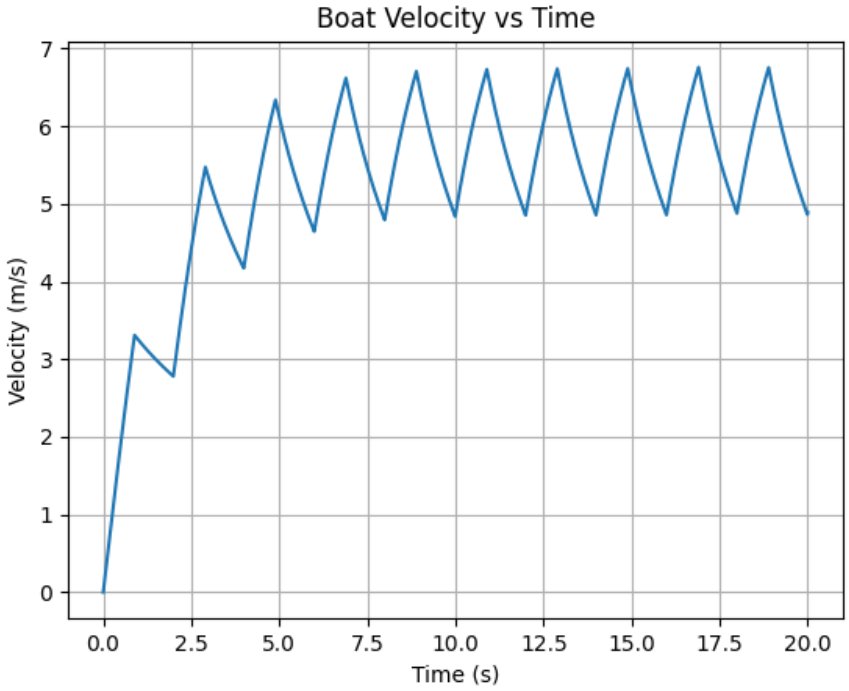


Figure 1: Baseline speed during the 20-second simulation.

6.673 m/s, but keeping the drive at 0.9 seconds also increases the drive fraction and input. This is therefore not an isolated comparison at constant work or power.

The model excludes blade slip, rower motion relative to the shell, varying force profiles, wind and waves. It has not been calibrated or compared with measured trajectories. More realistic thrust and drag measurements, timestep checks and experimental validation would be needed before applying the numerical predictions to a crew.

4 Stroke cycle and energy accounting

With cycle period $T = 2$ s and drive duration $d = 0.9$ s, the prescribed force is F_d when $0 \leq t \bmod T < d$, and zero otherwise. The resulting duty fraction is $d/T = 0.45$. The force acts on the combined modeled mass; it is not a direct measurement of force at an oar handle or a physiological power estimate.

Multiplying the equation of motion by v and integrating gives the continuous energy balance $W - D = \Delta K$. The discrete calculation evaluates force and power at the beginning of each step; the squared velocity update introduces a step-dependent residual. A small residual is a necessary consistency check, but does not verify the drag coefficient or thrust model. Velocity is clipped at zero in the code, so an energy balance would also need to account for that operation if it became active. It is not active in the reported baseline.

5 Sensitivity profiles

Figures 2–4 show the complete transient profiles rather than only the summary averages. The mass comparison changes acceleration and stroke-to-stroke fluctuations as well as the finite-interval average. The 20-second averages do not prove that a periodic steady state has been reached. Lower drag changes the relationship between thrust, speed and work, so it is not a comparison at equal mechanical power. The higher-rate case uses a longer fraction of each cycle for the drive and therefore changes more than stroke frequency.

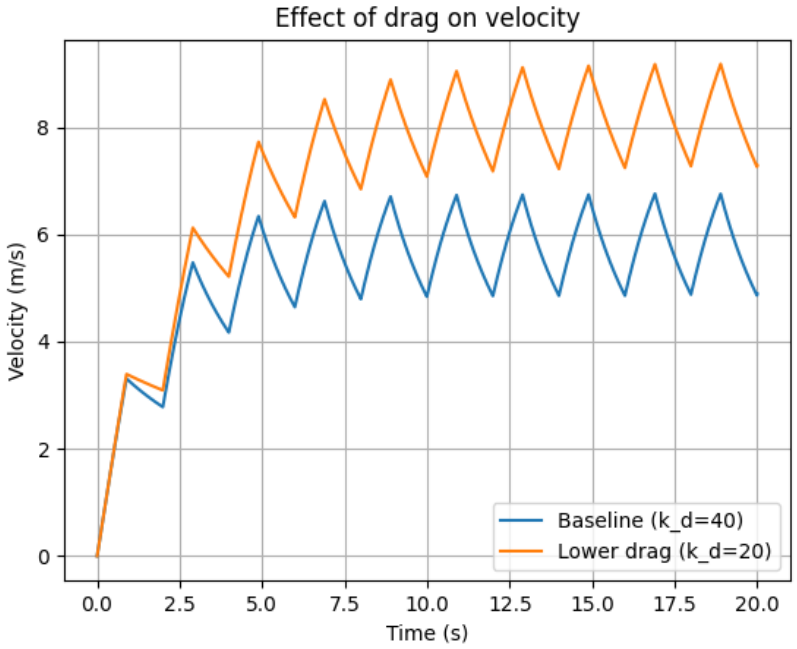


Figure 2: Velocity for baseline and lower drag.

6 Context and further work

Rowing involves moving body segments, oar-water interactions and hull motion that cannot be resolved by this single-mass model. A biomechanical review by Baudouin and Hawkins discusses performance factors and velocity fluctuations [1]; it provides context rather than validation of the present parameter choices. A useful next step is to fit time-resolved thrust and speed to independent measurements, assess parameter uncertainty, and compare predictions with held-out strokes. A longer simulation with a cycle-to-cycle convergence criterion would distinguish startup behavior from periodic operation. Comparisons of stroke rates should specify whether mean thrust, work per stroke, or average power is held fixed.

References

[1] A. Baudouin and D. Hawkins, "A biomechanical review of factors affecting rowing performance," *British Journal of Sports Medicine*, 36, 396–402, 2002. `https://bionics.seas.ucla.`

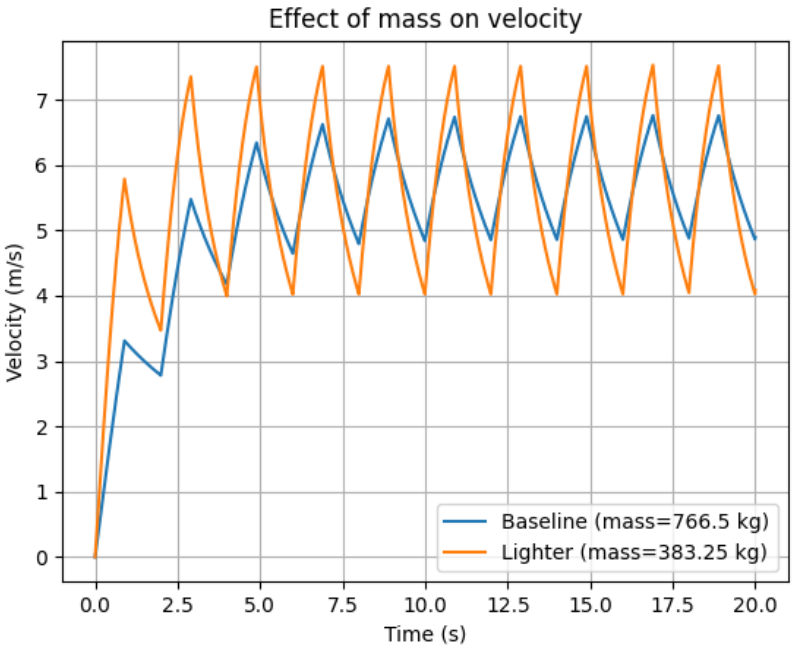


Figure 3: Velocity for baseline and lower mass.

`edu/education/Rowing/Math_Model_2002_01.pdf`.

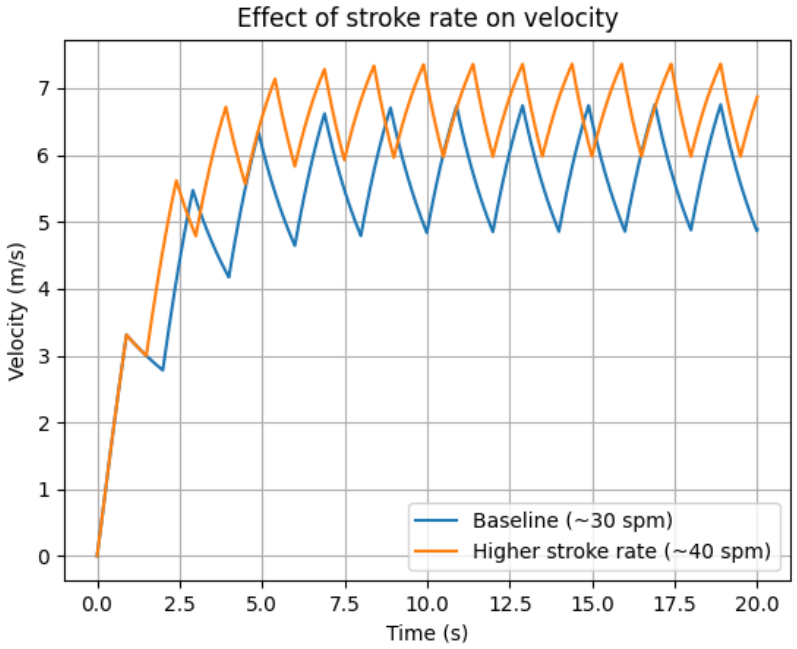


Figure 4: Velocity for two cycle periods with fixed drive duration.

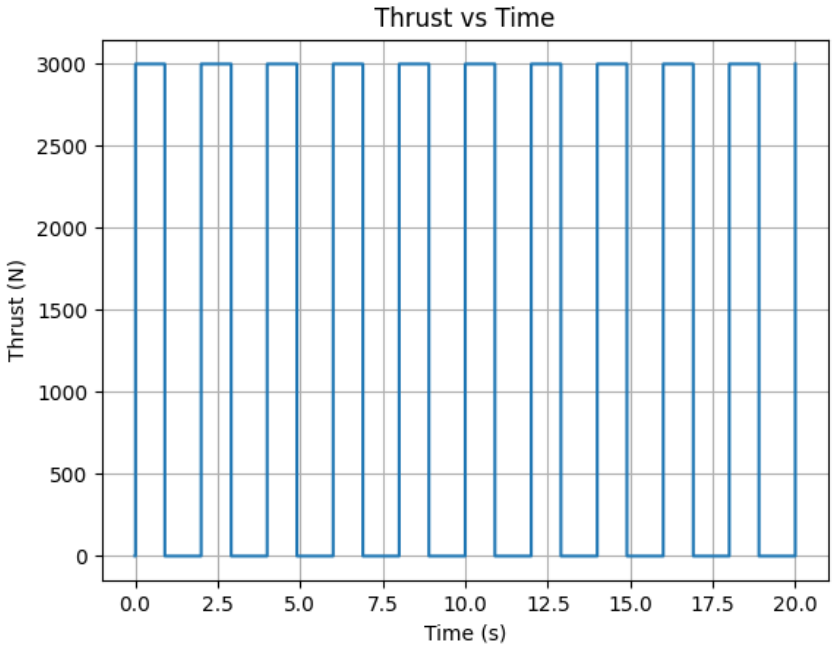


Figure 5: Prescribed thrust profile.